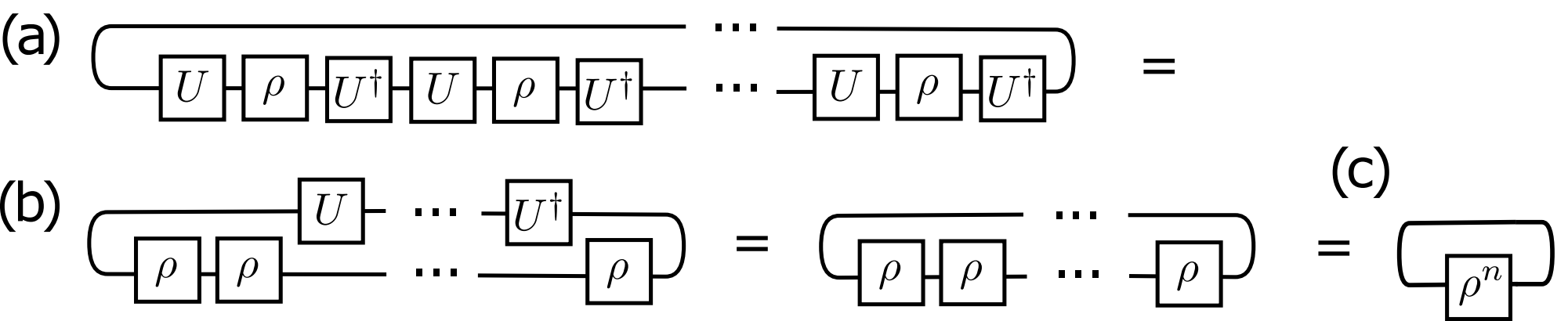
(a)
U
ρ
U†
U
ρ
U†
=
U
ρ
U†
(b)
U
U†
ρ
ρ
ρ
=
ρ
ρ
ρ
(c)
=
ρ^n